

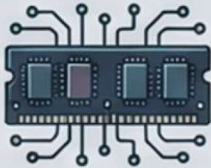
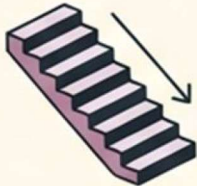
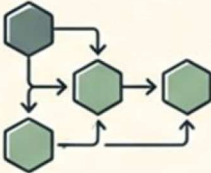
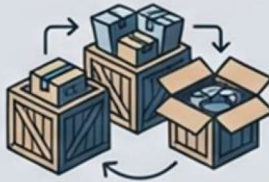


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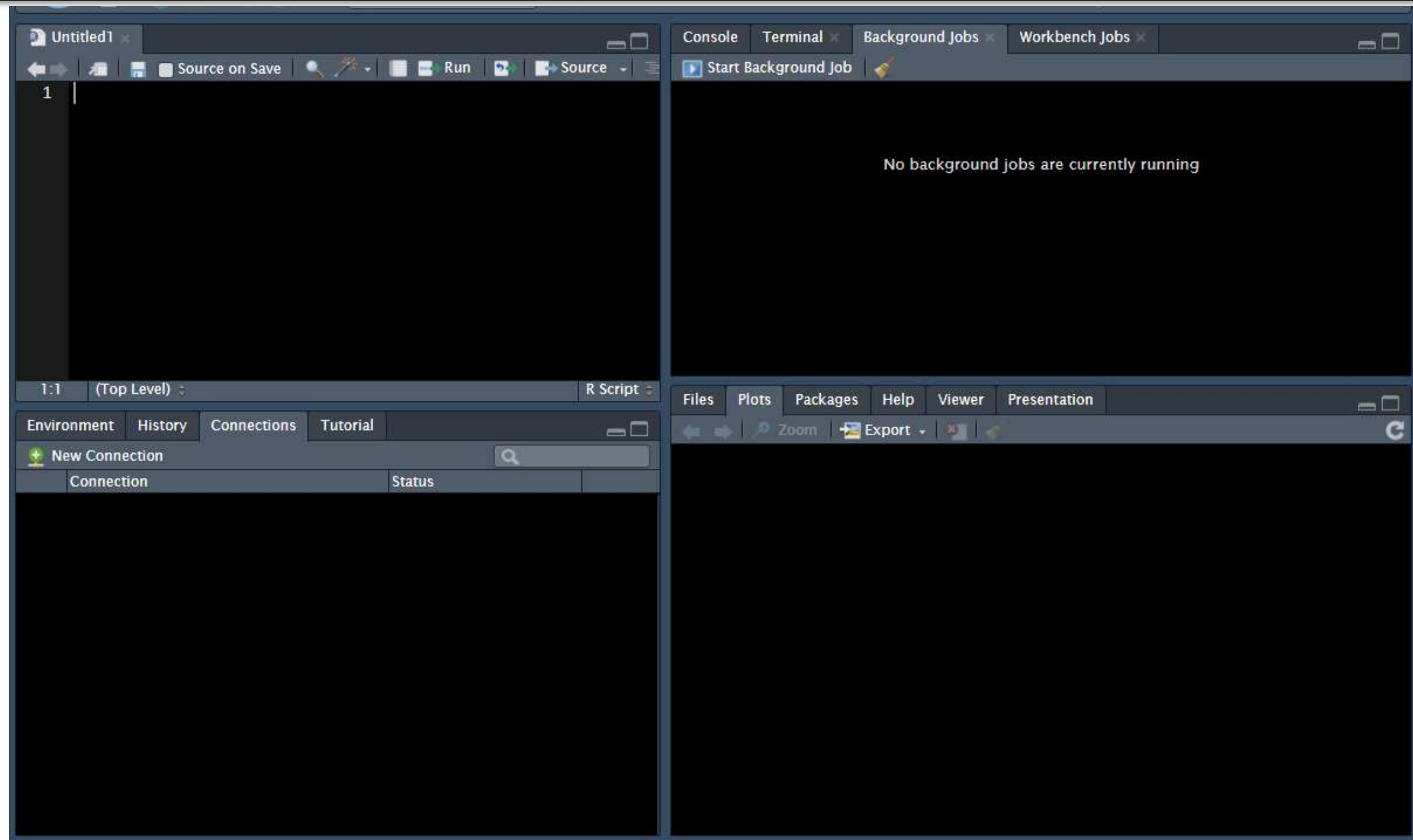
Jason Geslois, MPH, MAS, EMBA
Senior Epidemiologist

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Translating the Philosophy: SAS vs. R

Dimension	Legacy (SAS)	Modern (R)
Data Storage	 Disk-based tables. Persistent storage.	 In-memory Data Frames. Active RAM.
Execution	 Sequential PROCs & DATA steps.	 Interactive, object-oriented functions.
Ecosystem	 Proprietary, built-in modules.	 Open-source, community Packages.
Workspace	 Libraries and Libnames.	 Self-contained RStudio Projects.

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To protect and improve the health and environment of all Kansans

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The Command Center

 →  Translation Key

Source / Files

The Drafting Board.

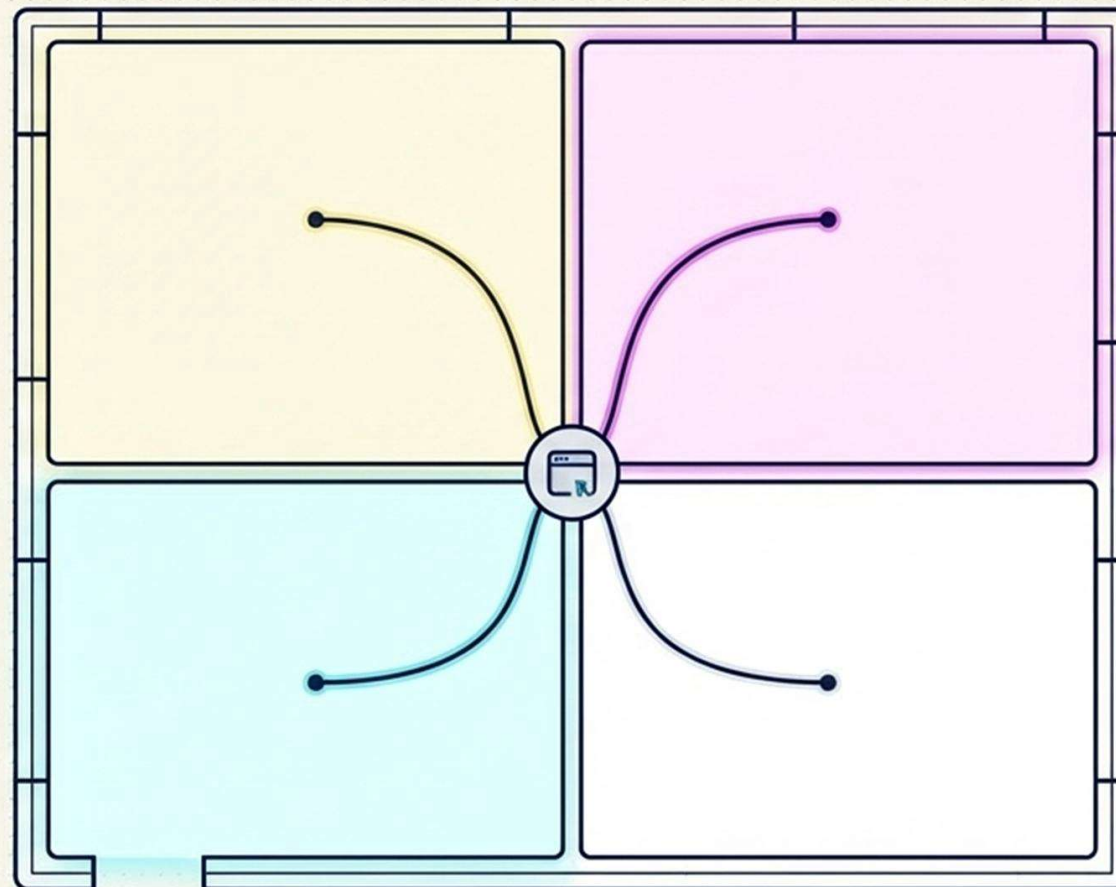
Where scripts, markdown, and quarto files are written.

Console

The Engine. Where code executes instantly.



Unlike SAS, you don't need a Run command for every line; the console is live.



Environment

Active Memory.

Displays loaded data and saved objects.

Output

The Display. Houses the Plot pane, Help pane, and local file directory.

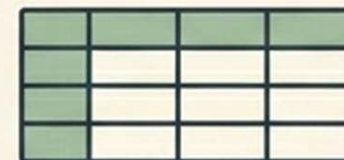
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The Drafting Table (Source)



Where you write reproducible code.

The Workbench (Environment)



View active objects, loaded Data Frames, and manage memory.

The Engine Room (Console)



For immediate execution and testing code on the fly.

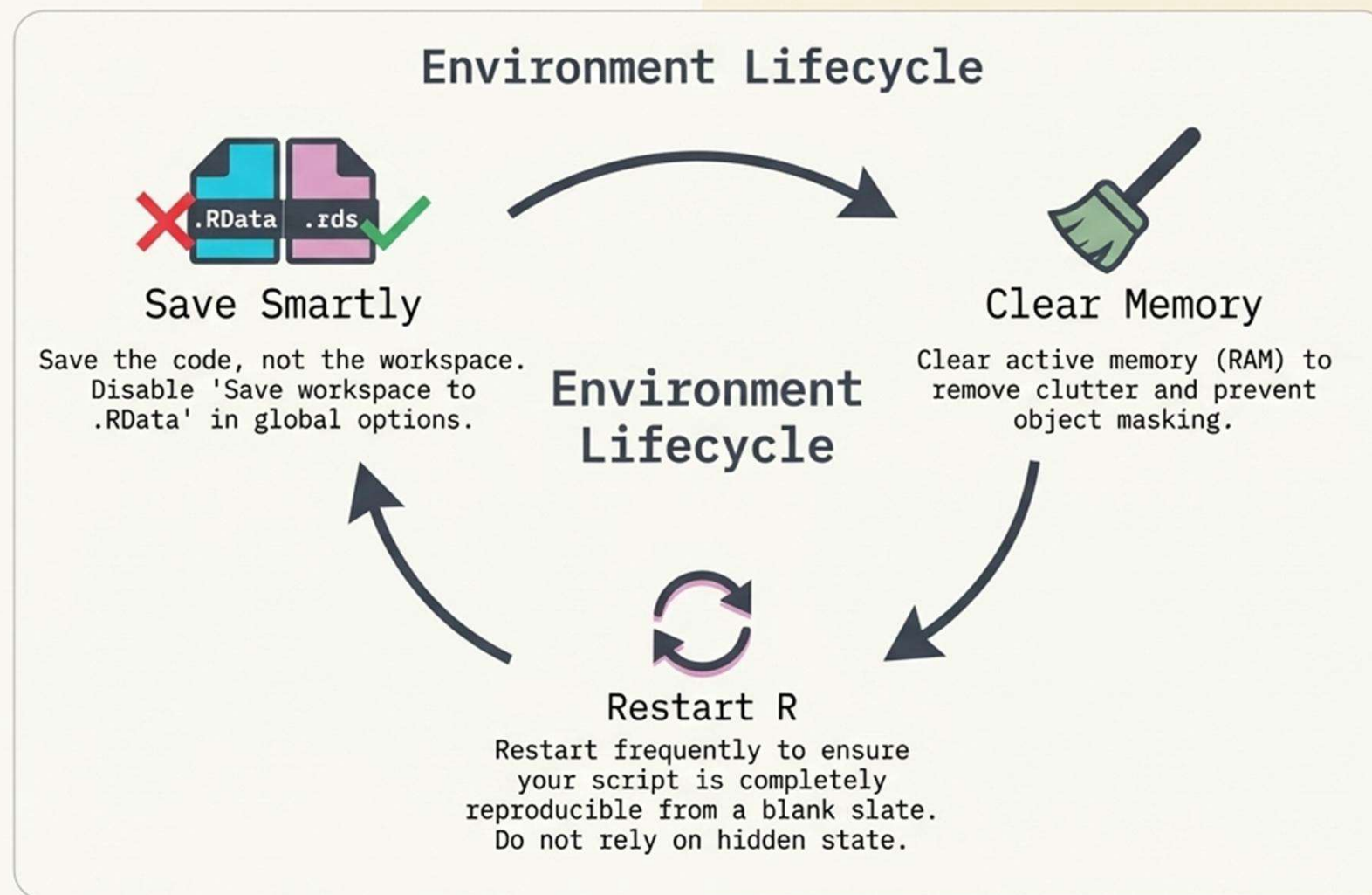
The Display (Output)



Review visualizations, read package documentation, and browse your directory.

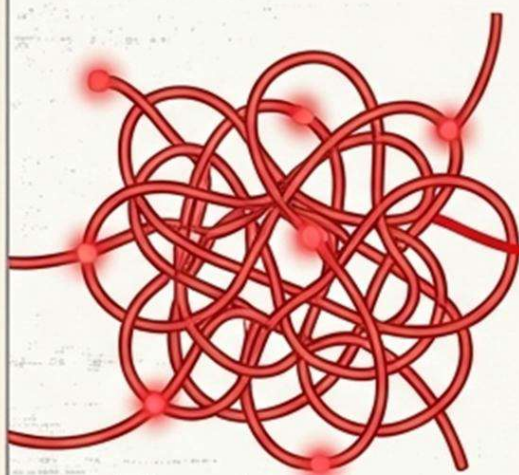
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Session Management and Active Memory



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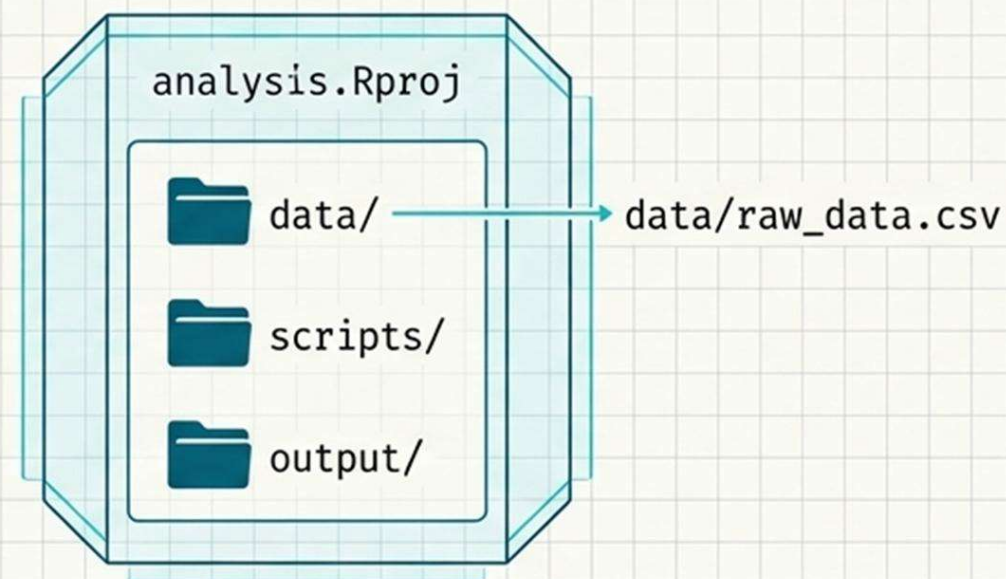
The Old Way



C:/Users/Name/
Documents/Project/
Data/raw_data.csv

Breaks immediately
if moved or shared
with a colleague.

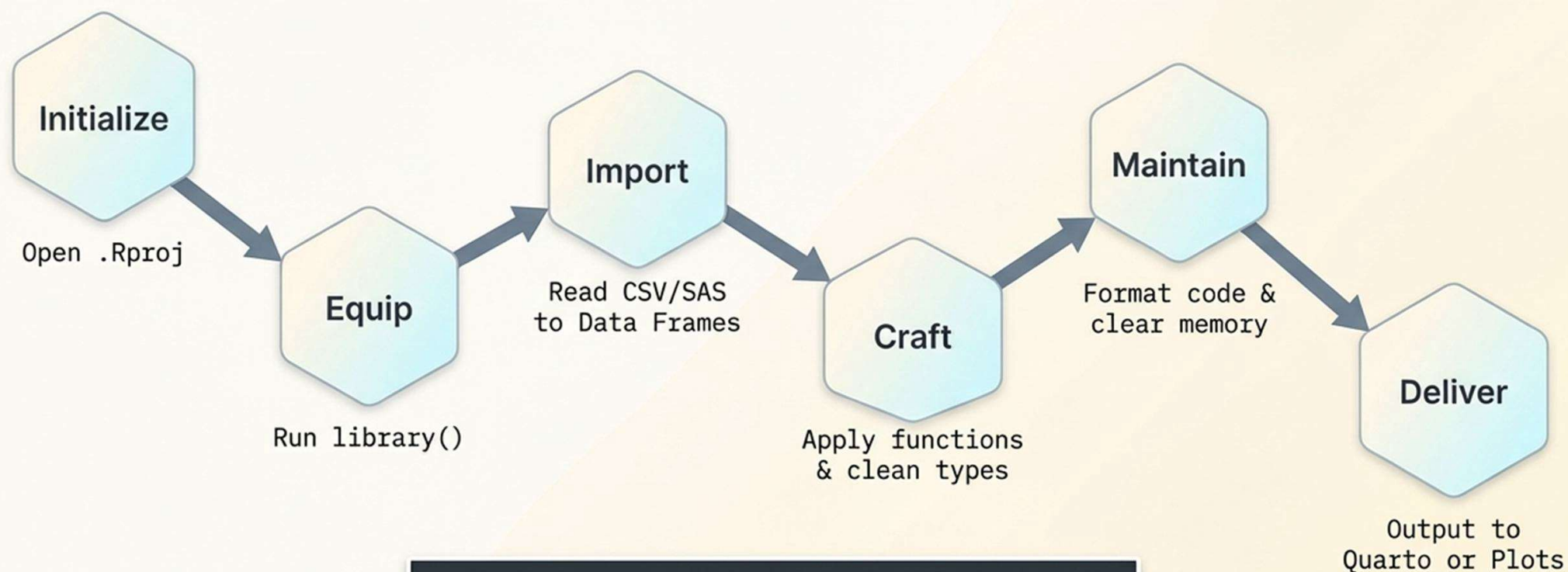
The R Project Way



A Project is a self-contained universe. By using relative paths, the code just works—no matter whose computer opens it.

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Synthesis: The R Workflow Blueprint



You now possess the blueprint. The transition from legacy scripts to reproducible environments begins here.

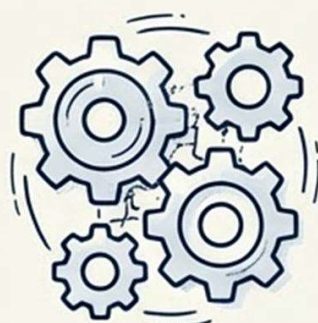
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Document Decision Matrix

Scripts (.R)

Use Case: Backend data processing, simple functions.

Analogy: The rough draft / engine room.



Markdown (.Rmd)

Use Case: Interleaving text, code, and plot panes.

Analogy: The standard analyst notebook.



Quarto (.qmd)

Use Case: Modern, multi-format publishing (HTML, PDF, slides).

Analogy: The final polished report.



The Modern Standard

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```
{r setup, include=FALSE}  
knitr::opts_chunk$set(echo = TRUE)
```

Loading Libraries & Data Directories

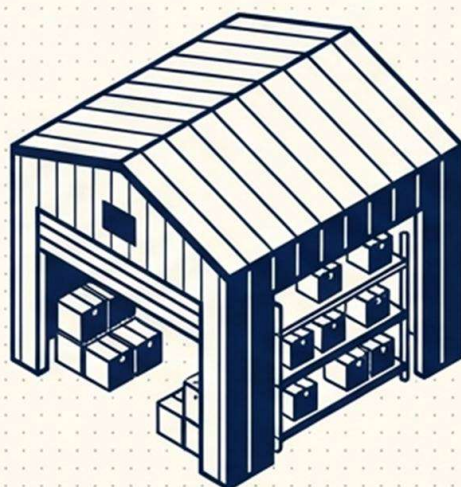
Here is where you first load the packages.

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The Package Supply Chain

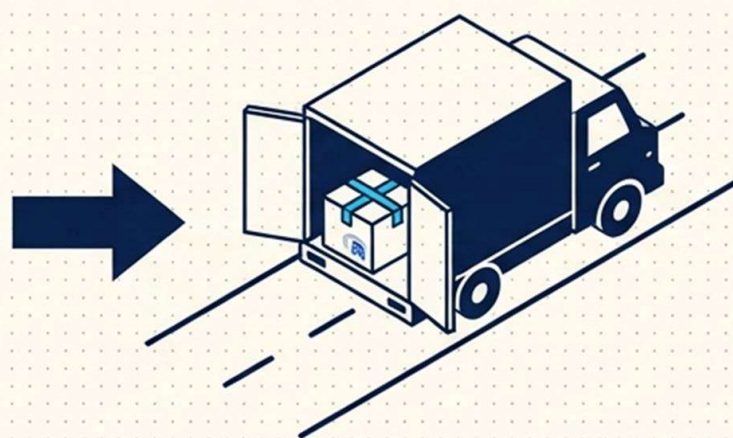
 SAS to R Translation Key 

CRAN (The Warehouse)



The global repository of open-source packages.

`install.packages()`



 **The Delivery:** Do this only once per machine to bring the tool to your hard drive.

`library()`



 **The Workbench:** Take the tool out of your local library and put it on your active desk. Do this every time you start a new session.

 NotebookLM

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Rainbow Code

```
(function(data){
  return data.filter(d =>
    (d.value > 10 &&
      (d.type === 'A' ||
        (d.type === 'B' && d.status === 'active'))));
})
```

Rainbow brackets prevent missing parenthesis errors and visually untangle complex nesting.

Find / Replace All

```
// unwe script anable
for (variable, seven's old_variable) = new_clean_var {
  return return, new_clean_var => old_variable;
}
//
if (new_variable = asetat) {
  repolve( new_clean_var variable, new_clean_var => new_clean_var));
}
```

Find: old_variable → Replace: new_clean_var

Instantly transform variable names across a whole script without hunting.

Status Bar & Global Options

Word Count: 1,420 Hidden Characters Markdown/Quarto

Word Count: 1,420 Line: 87, Col: 15

Enable word counts and hidden character views to optimize writing in Markdown and Quarto.

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Code Craftsmanship

SAS to R Translation Key 

The Messy Block

```
data mess; set raw;
  if x=1 then do;
    y=x*2;
    z=y+3;
    if z>10 then w=z-2; else w=z+2;
  end;
  else do;
    y=x*3;
    z=y+2; if z<5 then w=z/2;
  else w=z*2; end; run;
proc print data=mess; run;
```

The Clean Block

```
# Data Transformation for Final Output
messy_data <-
  raw_data |>
  mutate(
    y = case_when(
      x == 1 ~ x * 2,
      TRUE ~ x * 3
    ),
    z = case_when(
      x == 1 ~ y + 3,
      TRUE ~ y + 2
    ),
    w = case_when(
      z > 10 ~ z - 2,
      z < 5 ~ z / 2,
      TRUE ~ z + 2
    )
  )
```

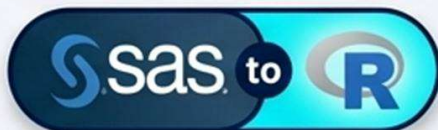
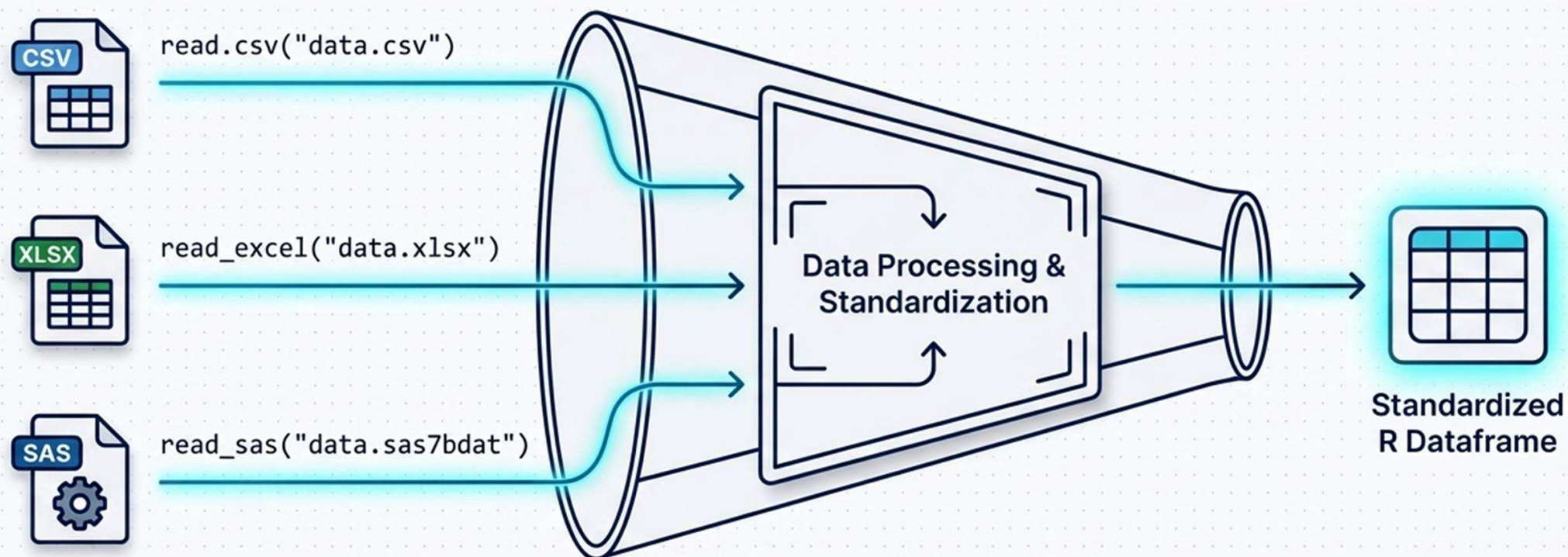
Commenting: Use # for inline context and to divide code into functional sections.

Reflow: Wrapping long text lines for readability.

Re-indent: Automatically aligning nested logic.

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Data Ingestion Pipeline



**SAS to R
Translation Key**

In SAS, you write LIBNAME statements.
In R, you assign reading functions directly to an object.

Anatomy of R Syntax



SAS

```
PROC MEANS DATA=mydata; VAR myvar; RUN;
```

R

```
summary_data <- calculate_mean(data, na.rm = TRUE)
```

The Environment Object

This pushes
the result into
memory.

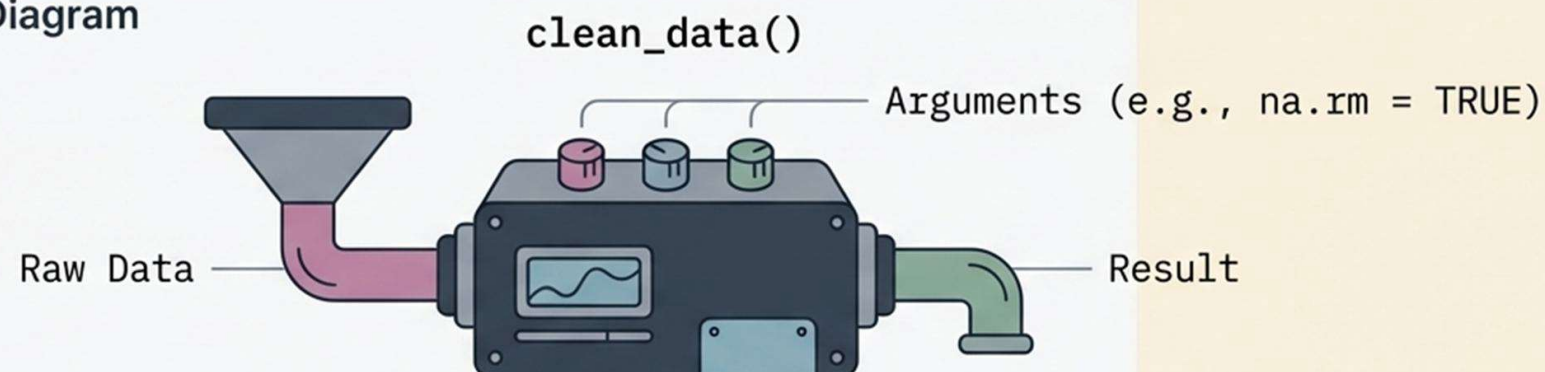
The Assignment
Operator

The Simple Function

The Arguments
(controlling function behavior)

Crafting with Functions

The Blueprint Diagram



Code Comparison

Run 1
`mean(sales_data, na.rm = FALSE)`

NA

Run 2
`mean(sales_data, na.rm = TRUE)`

42.5

Highlight and run different lines of code sequentially in the Console to instantly compare outputs and tweak function arguments.

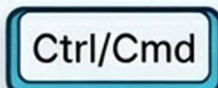
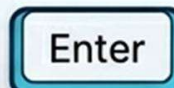
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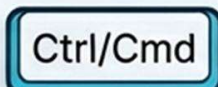
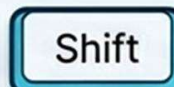
Quick Reference / Cheat Sheet

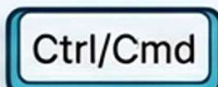
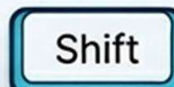
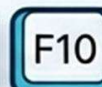


Top Keyboard Shortcuts

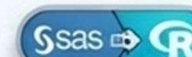
Assignment (<-)  

Run Code  

Find/Replace All   

Restart Session   

SAS to R Rosetta Stone Summary



Workspace → R Project

LIBNAME → read_*() functions

PROC → Modular Functions

.sas → .R, .Rmd, .qmd

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Keyboard Shortcut Quick Reference

Source on Save

Source Run Run & Debug Source

Ctrl+Shift+S Switch to Tab...
Ctrl+F12 Next Tab
Ctrl+F11 Previous Tab
Ctrl+Shift+F11 First Tab
Ctrl+Shift+F12 Last Tab

Source Navigation

Ctrl+F9 Back
Ctrl+F10 Forward
Ctrl+Alt+U Find Usages
Ctrl+F3 Use Selection for Find
Ctrl+F Find...
F3 Find Next
Shift+F3 Find Previous
Ctrl+Shift+J Replace and Find
Ctrl+. Go To File/Function...
Alt+Shift+G Go to Line...
Alt+Shift+J Jump To...
Ctrl+P Jump To Matching
Ctrl+Alt+Shift+E Expand To Matching
Ctrl+Shift+O Show Document Outline
Ctrl+Alt+Up Add Cursor Above Current Cursor
Ctrl+Alt+Down Add Cursor Below Current Cursor
Ctrl+Shift+Up Expand Selection
Ctrl+Shift+Down Shrink Selection
Ctrl+PgDn Go to Next Section
Ctrl+PgUp Go to Previous Section
Ctrl+Alt+A Split Into Lines
Ctrl+Alt+Shift+A Edit Lines from Start
Ctrl+Shift+F4 Use Visual Editor
Ctrl+Alt+Shift+Up Move active cursor up
Ctrl+Alt+Shift+Down Move active cursor down

Files

Ctrl+S Save
Ctrl+Alt+S Save All
Ctrl+Alt+Shift+N New R Script
Ctrl+O Open File...
Ctrl+W Close
Ctrl+Alt+Shift+W Close All Except Current
Ctrl+Shift+W Close All
Ctrl+Shift+F Find in Files...

Execute

Ctrl+Shift+S Source Active File
Ctrl+Shift+Enter Source with Echo
Ctrl+Shift+Enter Preview JS
Ctrl+Shift+Enter Preview SQL
Ctrl+Alt+G Source a File...
Ctrl+Alt+P Re-Run Previous
Ctrl+Enter Run Selected Line(s)
Alt+Enter Run Line(s) without moving cursor
Ctrl+Alt+R Run All

Console Terminal Background Jobs Workbench Jobs See All Shortcuts...

Start Background Job

Source Editor

Ctrl+Alt+I Insert Chunk
Ctrl+Shift+R Insert Section...
Ctrl+Alt+X Extract Function
Ctrl+Alt+V Extract Variable
Ctrl+Shift+C Comment/Uncomment Lines
Ctrl+I Reindent Lines
Ctrl+Shift+/ Reflow Comment
Ctrl+Shift+A Reformat Selection
Ctrl+Alt+Shift+D Show Diagnostics (Project)
Alt+L Collapse Fold
Alt+Shift+L Expand Fold
Alt+O Collapse All Folds
Alt+Shift+O Expand All Folds
Alt+Up Move Lines Up
Alt+Down Move Lines Down
Ctrl+D Delete Line
Ctrl+U Yank Line Up to Cursor
Ctrl+K Yank Line After Cursor
Ctrl+Y Insert Yanked Text
Alt+- Insert Assignment Operator
Ctrl+Shift+M Insert Pipe Operator
Ctrl+Alt+Shift+M Rename in Scope
Ctrl+Alt+Shift+R Insert Roxygen Skeleton
Ctrl+Backslash Request Copilot Code Suggestions

Source Control

Ctrl+Alt+D Diff Files
Ctrl+Alt+M Commit...

Build

No background jobs are currently running
Ctrl+Shift+K Compile PDF
Ctrl+Shift+K Preview
Ctrl+Shift+K Knit Document
Ctrl+Shift+K Render Document
Ctrl+Shift+K Run Document
Ctrl+Shift+B Install Package
Ctrl+Shift+L Load All
Ctrl+Shift+E Check Package
Ctrl+Alt+F7 Test Package
Ctrl+Shift+D Document

Files Plots Packages Help Viewer Presentation

Console

Ctrl+L Clear Console
Ctrl+Up Popup Command History

Zoom + Export

Terminal

Alt+Shift+R New Terminal
Alt+Shift+M Move Focus to Terminal
Alt+Shift+F11 Previous Terminal
Alt+Shift+F12 Next Terminal

Debug

Shift+F9 Toggle Breakpoint
F10 Execute Next Line
Shift+F4 Step Into Function
Shift+F7 Finish Function/Loop
Shift+F5 Continue
Shift+F8 Stop Debugging

Other

F1 Show Function Help
F2 Go To Function / File
Ctrl+Space Complete Code
Ctrl+Shift+F10 Restart R
Ctrl+Alt+F11 Previous Plot
Ctrl+Alt+F12 Next Plot
Ctrl+Shift+` Log focused element
Ctrl+Shift+W Change Directory

Accessibility

Alt+Shift+` Screen Reader Support

RStudio: Posit Software PBC

To protect and improve the health and environment of all Kansans

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Now for a Live Demo

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Thank You/Questions

